

San Jose State University

CS249 - Distributed Systems

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Project Proposal on

Creating a qBitTorrent-like application with persistent data storage.

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List of Abbreviations

IPFS Interplanetary File System

P2P Peer-to-Peer

DHT Distributed Hash Table

BEP BitTorrent Enhancement Proposal

Chapter 1: Introduction

1.1 Problem Statement

qBitTorrent is a common Peer-to-Peer ([P2P](#)) application that utilizes the libtorrent-rasterbar library for protocols and Qt toolkit for UI [insert qBitTorrent website citation here]. Libtorrent-rasterbar is a library developed by Arvid Norberg and is the protocol for BitTorrent and qBitTorrent [create citation for <https://github.com/airium/libtorrent-rasterbar>insert]. New features for the protocol are proposed in the form of a BitTorrent Enhancement Proposal ([BEP](#)), similar to that of PEP or Python Enhancement Proposal. For example, BEP 5 introduced Distributed Hash Table ([DHT](#)) protocol, BEP 11 introduced Peer Exchange, and BEP 52 introduced SHA-256 based infohashes and Merkle tree hashing.

Chapter 2: Research and Approach

2.1 Research Objective

The goal of this project is to analyze how bots have been detected in previous studies, implement some of those detection methods to the latest and largest dataset, Twibot-22, regarding X posts and accounts, and create our own bot in an attempt to avoid such detection methods to provide better insight as to how big of a problem bots really are.

It is important to note that creating a bot, analyzing the motivations behind creating bots, and following bot activity may involve ethical lapses such as the spread of misinformation. This can be due to many factors, such as deliberately spreading misinformation to push a narrative, known as disinformation. Another factor that is unavoidable, at least in terms of this project, is the risk that a GPT “hallucinates” and returns false information. A hallucination with respect to AI means that the information was presented as fact, even though the information is false. This was a very common occurrence in earlier GPT models, and although hallucinations have seen a significant downward trend since then, hallucinations can and still occur. Another ethical concern is regarding the use of social media apps and the privacy of its users’. Users interact on X on a daily basis and some of them have public accounts. It is important to maintain and respect the user’s privacy, regardless if their accounts are public or not.

2.2 Technical Approach

We will survey a variety of different methods that have been used for bot detection before, such as ML classifiers [7], LLM Bot Guard [4], and **NLP!** (NLP!) techniques [8]. We’ll organize each method based on what detected best and when the detection occurred. Afterwards, we will use the detection methods on the latest dataset, Twibot-22, which is a GitHub repository created by other researchers who are specializing in the field of bot detection [9]. Once that data is consolidated, we

will create our own X bot using the developer options that X allows its users and create an account from there. We will investigate methods on how to automate via GPTs since they now pass the Turing test and can imitate human responses the best. Afterwards, we will revisit the different methods of bot detection and test them against our own bot to see if it circumvents detection and report our findings.

Chapter 3: Requirements and Timeline

3.1 Requirements

For this project, we will need access to LLMs, MLs, and NLP techniques that have been used for bot detection. Some of these models are free to use, such as GPT-4.5 from OpenAI but others will require licenses or subscriptions. We will require access to the Twibot-22 which is given to us free of charge via the researcher's GitHub repository. And lastly, in order to have better engagement metrics on the bot account for X, a subscription for X will be required as many suspected bot accounts are also verified for a boost to engagement.

3.2 Timeline

The timeline for this project will be done over a period of 16 weeks, from the months of August to December of 2026. Since there are four parts to the project, we will split them up into four-week blocks.

The first set four weeks will focus on surveying the different methods and organizing based on best detection element. The second set four weeks will focus on applying those detection methods to the latest dataset, Twibot-22. This period will also involve setting up the environment to properly test on the dataset. The third set of four weeks will involve creating a bot via a GPT, specifically within the first two weeks of this block, and allowing it to interact with the platform for the remaining two weeks. For the last set of four weeks, while still collecting data from our bot, we will apply the methods of bot detection against our own bot with the data it generated. A Gantt chart is presented below to illustrate the timeline.

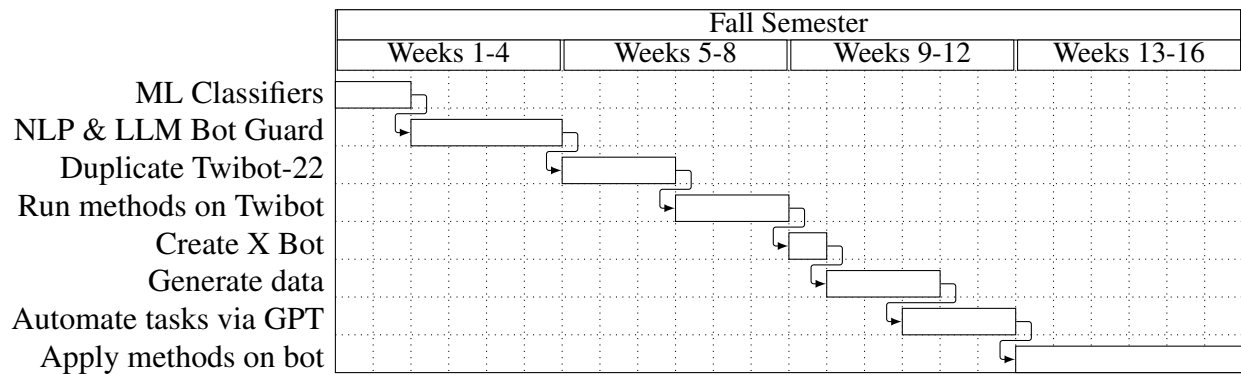


Figure 3.1: Gantt Chart

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